

An Introduction to Climate Modeling

Milestone 1 - Geography and Visualization

1 Read-in and Plot Geography

In this milestone, the goal is to read-in the geography information of planet Earth and plot it as a two-dimensional world map. We consider an equirectangular grid of the Earth, i.e., an equidistant rectangular grid in spherical coordinates, where the grid point (i, j) has the spherical coordinates (θ_i, φ_j) , where θ_i is the latitude between -90° south and 90° north (including the poles) and φ_j the longitude between -180° west and 180° east. The basis for this is the input file `The_World65x128.dat`, which describes the distribution of the different Earth surface types. This file contains a matrix $G \in \mathbb{N}^{65 \times 128}$ with entries $g_{ij} \in \{1, 2, 3, 5\}$, where the entry g_{ij} stores the Earth surface type at grid point (i, j) . Here, 1 represents the Earth surface type *land*, 2 represents *sea ice*, 3 represents *snow*, and 5 represents *ocean*. The grid resolution in longitude and latitude direction is 2.8125° . The basis for this distribution and grid is from Zhuang et al.¹ You can proceed as follows:

1. Write a function `read_geography`, which reads the file `The_World65x128.dat` from the folder `input` and outputs a matrix $T \in \mathbb{N}^{65 \times 128}$ with the classification of the earth surface types.
2. Write a function `robinson_projection`, which maps an equirectangular grid in spherical coordinates to the plane. For simplicity use the approximate formula by Beineke for the Robinson projection,

$$x(\theta, \varphi) = \frac{\varphi}{\pi} (0.0379\theta^6 - 0.15\theta^4 - 0.367\theta^2 + 2.666),$$
$$y(\theta, \varphi) = 0.96047\theta - 0.00857 \operatorname{sign}(\theta) |\theta|^{6.41},$$

where φ is the longitude and θ the latitude in radians. This function should return two matrices $X = x_{ij}$ and $Y = y_{ij}$, where $x_{ij} = x(\theta_i, \varphi_j)$, $y_{ij} = y(\theta_i, \varphi_j)$.

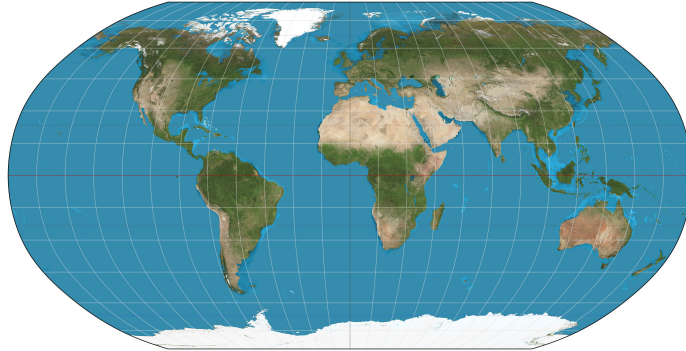


Figure 1: Robinson projection of the world ²

3. Write a function `plot_geo` that creates a plot of the Earth surface type g_{ij} against the mapped coordinates (x_{ij}, y_{ij}) .
4. Use these functions in a program and run it to check your results.

¹K. Zhuang, G.R. North, M.J. Stevens, *A NetCDF version of the two-dimensional energy balance model based on the full multigrid algorithm*, SoftwareX, Vol. 6, pp. 198-202, July 7, 2017.

²Daniel R. Strebe, https://en.wikipedia.org/wiki/File:Robinson_projection_SW.jpg

2 Bonus

1. Implement the Winkel tripel projection, as an alternative to the Robinson projection, given by

$$\begin{aligned}x(\theta, \varphi) &= \frac{1}{2} \left(\varphi \cos(\theta_1) + \frac{2 \cos(\theta) \sin(0.5\varphi)}{\text{sinc}(\alpha)} \right), \\y(\theta, \varphi) &= \frac{1}{2} \left(\theta + \frac{\sin(\theta)}{\text{sinc}(\alpha)} \right), \\ \alpha(\theta, \varphi) &= \arccos(\cos(\theta) \cos(0.5\varphi)),\end{aligned}$$

where θ_1 is the standard parallel and sinc is the unnormalized sine function that are

$$\theta_1 = \arccos\left(\frac{2}{\pi}\right) \quad \text{and} \quad \text{sinc}(x) = \begin{cases} \frac{\sin(x)}{x} & x \neq 0 \\ 1 & x = 0 \end{cases},$$

respectively.

2. Write a function to read in other data sets, like ICE-5G³, originally presented and analyzed by Peltier⁴. This function should parse this (binary) data which is provided in netCDF format. The data is given on an enriched grid, e.g., with 1° resolution. So, this read-in routine should also have the option to downsample the data in some way, e.g., via interpolation. After such downsampling, it can categorize the Earth surface types $g_{ij} \in \{1; 2; 3; 5\}$ based on RGB color values for plotting.

³<https://www.atmosp.physics.utoronto.ca/peltier/data.php>

⁴Peltier, W.R., Global glacial isostasy and the surface of the ice-age Earth: the ICE-5G (VM2) model and GRACE. Annual Review of Earth and Planetary Sciences, May 2004, Vol 32, 111-149.

3 Control Solutions

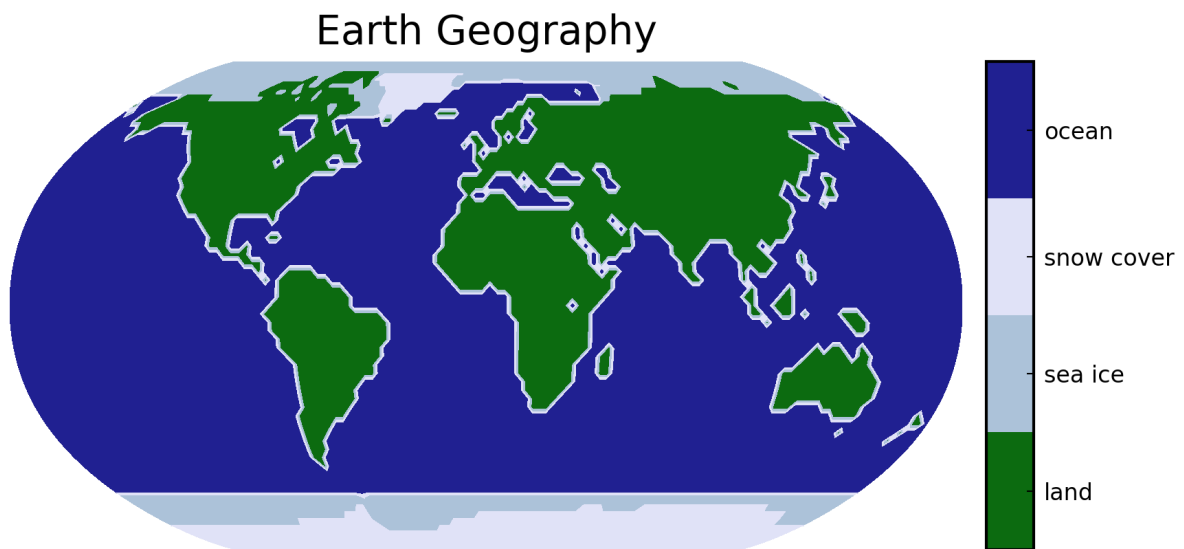


Figure 2: Robinson projection output